

Francesco Sammarco

Applied AI Engineer & Medical Imaging Researcher

ing.sammarco.francesco@gmail.com | [LinkedIn](#) | [Github](#) | [Personal Website](#)

PROFILE

AI/ML engineer and consultant with approximately 5 years of experience delivering customer-facing software, machine-learning solutions in Computer Vision and NLP field, advanced analytics platforms, and cloud infrastructure. Has designed and deployed end-to-end AI solutions across structured and unstructured data, including enterprise knowledge systems, agentic analytics, medical imaging, recommendation, anomaly detection, and optimisation.

WORK EXPERIENCE

AI/ML Engineer Consultant

Live Reply

Mar. 2026 – Present

Remote / Turin, IT

- Delivered a production GraphRAG knowledge system for approximately 100 users across 100+ technical documents, Git repositories, and SQL data; reduced operator search time by 52%.
- Developing an Azure analytics platform that turns transactional data into validated reports using SQL, object storage, specialized agents, workflow memory, and automated quality checks.
- Built Python and LangGraph systems on AWS to automate client business processes and provisioned Terraform environments for a real-time voice-agent system.

Applied AI Researcher

University of Tübingen

Sep. 2024 – Present

Remote / Tübingen, DE

- Built a reproducible PyTorch and SLURM pipeline for distributed training, inference, experiment tracking, and explainability evaluation across approximately 7,000 DICOM/NIfTI T1-weighted brain MRI scans.
- Developed explainability methods for dementia prediction from MRI, combining diffusion/flow-matching unconditional and conditional counterfactual generation from incomplete brain scans with saliency attribution.
- Contributed to [MedSAMix](#), a training-free medical-image segmentation model-merging project, and research on neuroimaging explainability.

Applied AI Researcher, Short-Term Collaboration

BeyondShape

February 2026 – March 2026

Naples, Italy

- Delivered two production AI pipelines for 3D spine reconstruction (based on gradient boosting, convolutional networks, generative methods) and ISIC dermatology anomaly detection (isolation forest on learnt embeddings).
- Worked on methods for pose normalisation based on deep learning based techniques (mainly using SMPL based models with Iterative Cloud Points techniques).
- Collaborated on the creation of a pipeline for mesh reconstruction from 2D images collected by high resolution cameras.

DevOps Engineer, Telecommunications Consultant

Hewlett Packard Enterprise (Swisscom)

2022 – 2024

Milan, Italy

- Built a Jenkins, Java, Spring, and SQL deployment system with containerized backend integrations for Swisscom, cutting release execution time by 89% and improving reliability.
- Coordinated as a project manager client, development, and operations teams to turn requirements into technical designs and production releases for the evolution of a major client platform, using AGILE methodology.

EDUCATION

Politecnico di Milano

Master of Computer Science and Engineering (ENG) | 110 cum laude/110, GPA: 29.43/30

Sep. 2020 – Oct. 2022

Milan, IT

- Artificial Intelligence focused path, thesis on Reinforcement Learning.

Università degli Studi di Napoli Federico II

Bachelor of Computer Science and Engineering (IT) | 110 cum laude/110, GPA: 29.70/30

Sep. 2017 – Jul. 2020

Naples, IT

- Background in computer science, mathematics, and physics (including databases, software engineering, and cybersecurity).

SELECTED ACHIEVEMENTS

ACM RecSys Challenge 2022

Mar. 2022 – Jul 2022

- Ranked 29th out of 250+ teams in the Dressipi session-based recommendation challenge, contributing ranking models and seasonality features later published in the challenge proceedings.

AI Summer School Instructor

Sep. 2025

- Designed and delivered hands-on AI teaching sessions for health and life-science participants, translating neuroimaging ML concepts into practical exercises with reproducible [code](#) [data](#).

Apple Academy

Sep. 2019 – Aug. 2020

- Ranked among top 100 and accessed a one year internship focused on app development in Swift and SwiftUI and project management fundamentals.

Cyber Challenge

May 2018 – Jul. 2018

- Ranked among top 50 and accessed closed number course focused on cybersecurity fundamentals, with execution of challenges on networking, reverse engineering, pwning and cryptography.

PROJECTS

Thesis: Non-Stationary Reinforcement Learning for Trading

Sep. 2021 – Oct. 2022

Title: Non-Stationary RL for Forex Trading with Automatic Market Regime Clustering

- Built a Forex trading system using [FQI](#) reinforcement learning, Gym, market-regime clustering, and XGBoost time-series classification.

Recommender System Challenges (RecSys 2022, H&M, Internal)

Mar. 2022 – Jul 2022

- Built media and fashion recommenders with collaborative filtering, matrix factorization, SLIM, GRU4Rec, and LightGBM on AWS EC2; RecSys 2022 solution ranked 29th of 250+ teams (0.1845 private MRR). [Code](#); paper in [Publications](#).

Artificial Neural Network Challenges

Oct 2020 – Jan 2021

- Built YOLO and RNN pipelines for image detection, segmentation, text classification, and sequence modeling; placed top five in 2 of 3 [Kaggle challenges](#).

Thesis on Genetic Algorithms

Apr. 2020 – Jul. 2020

Title: Application of Genetic Algorithms on Flexible Job Shop Scheduling problem

- Built a Python genetic-algorithm optimizer for flexible job-shop scheduling, improving production planning by jointly optimizing machine assignments and job order. [Code](#).

PUBLICATIONS

Yang Y.*, Su G.*, Hu J.*, Sammarco F., Geiping J., Wolfers T., *Medsamix: A training-free model merging approach for medical image segmentation.*, arXiv preprint arXiv:2508.11032 (AAAI), (2025) - [DOI](#)

Della Volpe N.*, Mainetti L.*, Martignetti A.*, Menta A.*, Pala R.*, Polvanesi G.*, Sammarco F.*, Maurera F. B. P., Bernardis C., and Dacrema M. F., *Lightweight Model for Session-Based Recommender Systems with Seasonality Information in the Fashion Domain.*, In Proceedings of the Recommender Systems Challenge, (2022) - [DOI](#)

TECHNICAL SKILLS

Programming Languages: Python, C/C++, MATLAB, SQL, Java, Cython, JavaScript, Swift, C#, HTML/CSS

ML/Data Libraries: PyTorch, OpenCV, TensorFlow, scikit-learn, nilearn, pandas, NumPy, matplotlib

Frameworks & Tools: SLURM, Docker, Kubernetes, Git, GitHub/GitLab, AWS, MLflow, Weights & Biases, Spring, Jenkins

Languages: *Italian:* C2 (native), *English:* C1, *German:* A2 (studying)